Practice Set 11 Paper 1 – Basic Sciences in Urology

1. Q1. Describe pzthophysiology of urosepsis and its management.
2. Q2. Describe anatomy of pelvic floor in females and its importance in surgeries described for SUI
3. Q3a. Retrocaval ureter
4. Q3b. Glandular hypospadias
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Practice paper compiled from the question bank.

Q1. Describe pzthophysiology of urosepsis and its management.

Type: LAQ (25 marks) (also answered in 2017-Summer Paper 1, Q2)

"Pzthophysiology" is read as pathophysiology.

Definitions

- **Urosepsis:** sepsis whose source is the urinary tract or male genital organs. It is usually Gram-negative and usually follows obstruction or instrumentation.

Term (Sepsis-3, 2016)	Definition
Sepsis	Life-threatening organ dysfunction from a dysregulated host response to infection: acute rise in **SOFA (Sequential Organ Failure Assessment) score ≥2**
qSOFA (quick SOFA)	Respiratory rate ≥22/min, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk of death
Septic shock	Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite adequate fluids

- SIRS (systemic inflammatory response syndrome) criteria were dropped from the definition as non-specific.

Causes and risk factors

- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy, giving infected hydronephrosis or pyonephrosis.
- **Parenchymal infection:** emphysematous pyelonephritis, renal or perinephric abscess, prostatic abscess, Fournier gangrene.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy at high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy (sepsis 0.3–3.1%), catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.
- **Organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.

Pathophysiology

1. Bacteria and endotoxin enter the blood

- Obstruction causes stasis and raises pelvic pressure. Bacteria and endotoxin then enter the circulation by **pyelovenous and pyelolymphatic backflow**. High irrigation pressure at endoscopy does the same.
- Fragmenting infected stones releases endotoxin. Stone culture predicts postoperative sepsis better than bladder urine culture.
- An obstructed infected kidney behaves like an **abscess**: antibiotics barely reach it, and bladder urine may be sterile.

2. Host response

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Endotoxin (LPS, lipopolysaccharide) or Gram-positive wall products
  → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB)
  → TNF-α (tumour necrosis factor α), IL-1, IL-6 (interleukins)
  → complement, coagulation and endothelial activation
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- **Vasodilatation:** inducible nitric oxide lowers systemic vascular resistance ("warm" shock).
- **Capillary leak:** endothelial glycocalyx damage causes hypovolaemia and oedema.

- **Coagulopathy:** tissue factor release causes microthrombi and DIC (disseminated intravascular coagulation).
- **Myocardial depression and mitochondrial failure:** cells cannot use oxygen, so lactate rises.
- **Immunoparalysis:** a later anti-inflammatory phase invites secondary infection.

3. Organ dysfunction: AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver failure, DIC, multi-organ failure.

Clinical features

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often begins within hours of endourology.

Investigations

- **Lactate**, two blood cultures, urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, blood gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas or abscess.

Management

A. Resuscitation and antibiotics (SSC, Surviving Sepsis Campaign 2021)

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Recognise (SSC: do not use qSOFA alone for screening)
 → lactate + cultures → IV (intravenous) antibiotics within 1 hour
 → 30 mL/kg balanced crystalloid within 3 h if hypotensive or hypoperfused
 → noradrenaline first line, target MAP 65; add vasopressin
 → persistent vasopressor need: hydrocortisone 200 mg/day
 → ICU (intensive care unit); repeat lactate, capillary refill
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- **Empirical antibiotics** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

B. Source control: the key urological step

- **Drain the obstructed infected kidney urgently** by retrograde **double-J stent** or **PCN (percutaneous nephrostomy)**. Equally effective in a randomised trial (Pearle, 1998).
  - Stent: suits coagulopathy; avoid high-pressure contrast injection.
  - PCN: suits pyonephrosis or large impacted stones; gives pelvic urine for culture.
- **No definitive stone surgery during sepsis**; treat the stone once infection clears.
- Other sources: drain abscesses; emphysematous pyelonephritis (percutaneous drainage; nephrectomy for diffuse gas without fluid); prostatic abscess drainage; change the catheter; debride Fournier gangrene.

C. Supportive care

- Oxygen or ventilation; insulin if glucose ≥180 mg/dL; transfuse below haemoglobin 7 g/dL; thromboprophylaxis; renal replacement if needed.

D. De-escalation

- Narrow antibiotics to culture at 48–72 h; switch to oral when stable. Usual course 7–14 days; shorter courses suffice after good source control.

Special situations and prevention

- **Also know:** pregnancy (stent or nephrostomy, avoid fluoroquinolones); fungal urosepsis (remove catheter, antifungal, drain fungal balls).

- **Prevention:** sterile urine before elective endourology; prophylaxis; low irrigation pressure and access sheath; short operating time; stage the procedure if pus is found.
- **Prognosis:** better than sepsis from other sites when drained early; delayed drainage is the main avoidable cause of death.

Diagram to draw: obstructed kidney → pyelovenous backflow → cytokine cascade → shock and organ failure.

> **Key points to remember:**
> - Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
> - Stent = nephrostomy for drainage; delay definitive stone surgery.

Q2. Describe anatomy of pelvic floor in females and its importance in surgeries described for SUI

Type: LAQ (25 marks) (pelvic floor anatomy also answered in 2017-Winter Paper 3, Q2 and Practice Set 2 Paper 1, Q1)

Introduction

- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- Pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**, anchored to the bony pelvis. Every SUI operation uses one of these anchors.

A. Anatomy of the female pelvic floor

1. Pelvic diaphragm (levator ani + coccygeus)

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal sling
Pubococcygeus (pubovaginal part)	Posterior pubis and arcus tendineus → viscera, anococcygeal raphe	Slings urethra and vagina
Iliococcygeus	ATLA (arcus tendineus levator ani), ischial spine → raphe, coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine → sacrum, coccyx	Lies on sacrospinous ligament

- Nerve to levator ani **(S3–S5)** runs on the pelvic surface; **pudendal nerve (S2–S4)** supplies the sphincters.
- Tonic contraction closes the **urogenital hiatus**. Muscle fails first (birth injury, avulsion); the fascia then carries the whole load and stretches.

2. Endopelvic fascia and ligaments

- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to the ischial spine.
- **Pubourethral ligaments** suspend the urethra from the pubis; **urethropelvic ligaments** anchor it to the ATFP.
- **Cardinal–uterosacral complex** suspends the apex; **rectovaginal fascia** posteriorly.

3. Perineal membrane, perineal body and urethra

- Perineal membrane spans the pubic arch; urethra and vagina pierce it.
- **Striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; continence zone is the **proximal and mid-urethra**. The **oestrogen-dependent vascular submucosa** gives up to 30% of closure force.

4. DeLancey levels (1992)

Level	Structures	Defect
I suspension	Cardinal–uterosacral	Uterine or vault prolapse
II attachment	Paravaginal attachment to ATFP	Cystocele (central or paravaginal), rectocele
III fusion	Urethra, perineal membrane, perineal body	**Urethral hypermobility**, perineal descent

B. Theories of continence and the operations they produced

Theory	Mechanism	Surgical answer
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of the abdominal pressure zone	Restore position: Burch, MMK (Marshall–Marchetti–Krantz)
Hammock (DeLancey 1994)	Urethra is compressed against a firm backing of anterior vaginal wall attached to levator and ATFP; repositioning not essential	Restore backing
Integral theory (Petros and Ulmsten 1990)	Laxity of pubourethral ligaments at the **mid-urethra**	Tension-free mid-urethral sling
ISD (intrinsic sphincter deficiency; McGuire 1978)	Urethra fails to close; low VLPP (Valsalva leak point pressure) <60 cmH_2O	Sling, bulking, AUS (artificial urinary sphincter)

- Most SUI includes some ISD. So slings work in hypermobility, but Burch fails in ISD with a fixed urethra.

C. Importance in SUI surgery

Operation	Anatomy used	Anatomical hazards
Burch colposuspension	Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament**	Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina
Paravaginal repair / VOS (vagino-obturator shelf)	Lateral vaginal sulcus to **ATFP** or obturator fascia (level II)	Obturator vessels and nerve
Retropubic MUS (mid-urethral sling), TVT	Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline	Bladder perforation (4.5%), vessels, bowel
TOT (transobturator tape)	Through obturator membrane, **obturator internus and externus**; exit at adductor longus tendon notch	Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%)
SIMS (single-incision mini-sling)	Anchors in obturator internus or endopelvic fascia	Works by restricting mobility
Autologous PVS (pubovaginal sling)	Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus	Voiding dysfunction; best for ISD
Bulking agents	**Submucosa** coaptation	Repeated injections

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed (and kinked) during stress.
- **TOMUS (2010):** retropubic 80.8% vs transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.
- **Also know:** needle suspensions abandoned (sutures pulled through); anterior colporrhaphy is not an SUI operation; levator palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment; uncorrected level I loss causes recurrent prolapse after SUI repair.

Diagram to draw: cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

Key points to remember:

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey I suspension, II attachment (ATFP), III fusion (urethral hypermobility).
- Integral theory → mid-urethral sling; hammock → backing, not repositioning.
- Burch → Cooper's ligament; TOT → obturator foramen; PVS → bladder neck.

Q3a. Retrocaval ureter

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 3, Q3d)

Definition

- The **right ureter** passes **behind the IVC (inferior vena cava)**, then crosses in front of it from medial to lateral before resuming a normal course.
- Better named **"preureteral vena cava"**: the anomaly is venous; the ureter is trapped.

Embryology

- Fetal veins form a collar around each ureter: posterior cardinal and supracardinal (dorsal), subcardinal (ventral).
- Normally the right **supracardinal** vein, **dorsal** to the ureter, forms the infrarenal IVC.
- If the **lumbar subcardinal vein** (ventral) persists as the IVC, the ureter is trapped behind it.
- Variant: double right IVC, with the ureter between the limbs.

Incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.

Types (Bateson and Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower, L3–L4	Higher, at renal pelvis level
Shape	**J or "fishhook"**	Smooth horizontal curve
Obstruction	Yes, at iliopsoas edge	Mild or none

Investigations

- **CT urography (computed tomography urography): investigation of choice**; MRI (magnetic resonance imaging) alternative.
- IVU (intravenous urography): ureter not seen beyond the J hook. Retrograde: **S-curve** at L3–L4. **Diuretic renogram**: degree of obstruction, split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

Management

- **Non-obstructed (usually type II):** observe.
- **Obstructed:** divide the ureter, transpose it **anterior to the IVC**, and do a spatulated **ureteroureterostomy** (or ureteropyelostomy) over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect the ureteric blood supply.
- **Laparoscopic or robotic** (transperitoneal or retroperitoneal) now preferred. Nephrectomy if non-functioning.
- **Also know:** retroiliac ureter (L5–S1); bilateral cases in situs inversus.

Diagram to draw: right ureter hooking behind the IVC with a fishhook pelvis.

> **Key points to remember:**
> - Venous anomaly (persistent subcardinal vein); right side; 1 in 1500; male 3:1.
> - Type I fishhook obstructs; type II does not.
> - CT urography; anterior transposition and ureteroureterostomy.

Q3b. Glandular hypospadias

Type: SAQ (10 marks)

"Glandular" is read as glanular hypospadias: meatus on the glans.

Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.
- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

Operations

Procedure	Suitable for	Key point
Urethromeatoplasty	Stenotic meatus, blind pit	Heineke–Mikulicz: vertical incision, horizontal closure
MAGPI (meatal advancement and glanuloplasty, Duckett 1981)	Glanular meatus, mobile skin	Advance meatus, two-layer glanuloplasty; retracts if used more proximally
MIV (M inverted V) glansplasty	Glanular or coronal	Avoids MAGPI retraction
GAP (glans approximation)	Wide deep groove, fish-mouth meatus (MIP)	Tubularise and close glans wings
TIP (tubularised incised plate, Snodgrass 1994)	Narrow plate	Most used; slit-like meatus
Mathieu flip-flap; Thiersch–Duplay	Selected	Stenosis, fistula

- Preputioplasty or circumcision; dartos barrier layer.

Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

Diagram to draw: MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
>
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

Q3c. Treatment of submeatal stricture urethra in males

Type: SAQ (10 marks) (meatal stenosis also answered in 2019-Summer Paper 2, Q3a)

"Submeatal stricture" is read as a stricture of the meatus and fossa navicularis (glanular urethra).

Causes

- **LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]):** main inflammatory cause today; begins at the glans and meatus and progresses proximally.
- **Iatrogenic:** catheter or instrument trauma, TURP (transurethral resection of prostate).
- **Failed hypospadias repair**; post-circumcision ammoniacal meatitis (children).

Evaluation

- Spraying stream, straining; uroflowmetry.
- **RUG (retrograde urethrogram) before any meatal surgery**: a proximal stricture often coexists, especially in LS.
- Biopsy suspicious lesions: LS turns malignant (squamous cell carcinoma) in up to 8%.

Treatment

Option	Method	Comment
Topical steroid	Clobetasol 0.05% once or twice daily for 1–2 months, then alternate days	Early LS
Dilation or self-calibration	Meatal dilator lubricated with clobetasol	Elderly, unfit; seldom durable
Ventral meatotomy	Crush and incise ventrally; **three sutures** (apex and each side), mucosa to glans skin; dilate afterwards	Simple stenosis; in LS extend it with an inverted-V relaxing incision
Dorsal meatotomy (Y-V plasty)	Excise the scar ridge, advance a flap	Effective; cosmesis poorer
Fossa navicularis reconstruction (flaps)	**Jordan flap:** ventral urethrotomy through the stenosis; **transverse ventral penile skin island on dartos** transposed and inverted into the defect. **De Sy:** ventral longitudinal island advanced	Non-LS; excellent function and cosmesis
Graft	**BMG (buccal mucosal graft)** inlay or onlay, one or two stages; glans wings developed	**LS: oral mucosa, never genital skin**
Perineal urethrostomy	Bypass	Pan-anterior LS, failed repairs, elderly

- LS affects genital skin, so skin flaps recur; it spares the proximal urethra, so perineal urethrostomy works.
- Children: ventral meatotomy plus parental dilation for 7–10 days.

Complications

- Restenosis (commoner in LS), bleeding, spraying, poor cosmesis.

Diagram to draw: Jordan flap: transverse ventral skin island on dartos inverted into the ventral meatotomy.

Key points to remember:

- RUG first; adult meatal stenosis is usually the tip of a fossa stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Q3d. TOT in SUI

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 2, Q3f)

Introduction

- **TOT (transobturator tape):** a **MUS (mid-urethral sling)** of macroporous monofilament polypropylene passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

Mechanism

- **Integral theory:** pubourethral ligaments and the suburethral hammock maintain continence at the mid-urethra.
- A loose tape supports the posterior urethral wall, which is compressed against it on straining. It anchors in **obturator internus and externus**.

Indications and evaluation

- **SUI (stress urinary incontinence)** or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training).
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated. Counsel about permanent mesh and alternatives.

Technique (outside-in)

1. Lithotomy, hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm from the meatus; dissect to the ischiopubic ramus.
2. Groin stab incision at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal sulci.
5. Tension-free: scissors pass easily between tape and urethra.

- **Inside-out** (de Leval): fewer vaginal perforations, more voiding dysfunction.

Outcomes

- **TOMUS (2010):** objective success 77.7% vs 80.8% retropubic, not significantly different.
- Poorer with **ISD (intrinsic sphincter deficiency)** or a **fixed urethra**: retropubic sling preferred.

	TOT	Retropubic
Bladder perforation	0.6%	4.5%
Groin pain	6.4%	1.3%
Voiding dysfunction	Less	More
Long-term repeat surgery	Possibly more	Less

Complications

- Groin or thigh pain, vaginal mesh exposure, voiding dysfunction (sling incision), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- **Also know:** SIMS (single-incision mini-sling); alternatives: Burch colposuspension, autologous fascial sling, bulking agents.

Diagram to draw: pelvis with tape under the mid-urethra through both obturator foramina.

Key points to remember:

- Delorme 2001; anchors in obturator muscles.
- TOMUS: equivalent to retropubic (77.7% vs 80.8%).
- Fewer bladder injuries, more groin pain; weaker in ISD.

Q3e. Testicular biopsy

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 4, Q2)

Indications

- **Azoospermia** with **normal testicular size, palpable vasa and normal FSH (follicle-stimulating hormone)**: separates **OA (obstructive azoospermia)** from **NOA (non-obstructive azoospermia)** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Clues to obstruction: FSH ≤7.6 IU/L and testis long axis >4.6 cm.
- Because ICSI (intracytoplasmic sperm injection) needs one sperm, biopsy is **diagnostic and therapeutic**: plan it as **TESE (testicular sperm extraction)** and **cryopreserve** sperm.
- **Not needed:** CBAVD (congenital bilateral absence of the vas deferens) with normal FSH; high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb (azoospermia factor) deletion.

Types

Type	Method	Use	Limits
Open biopsy	Small albugineal incision, pea-sized sample	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core	14 G biopsy gun, cord block	Fresh retrieval in OA	Blind; epididymis or testicular artery injury
Aspiration (TESA, testicular sperm aspiration)	23 G needle, suction, 3–4 passes	OA; backup to epididymal retrieval	Poor architecture; low yield in NOA
Micro-TESE (microdissection)	Operating microscope, plump tubules	NOA	Long operation

Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep**, then wet squash prep for sperm; negative → contralateral biopsy.
7. Close tunica vaginalis separately; no antibiotics.

Histology

Pattern	Meaning
Normal spermatogenesis	Obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis	End-stage (e.g. Klinefelter)

Complications

- **Haematoma** (commonest); **inadvertent epididymal biopsy** (most serious, causes obstruction); infection rare.

Diagram to draw: open biopsy: small albugineal window with extruding tubules snipped off.

> **Key points to remember:**
> - Indication: azoospermia, normal testes, vasa and FSH.
> - Bouin's, never formalin; no forceps; touch prep.
> - Haematoma commonest; epididymal biopsy most serious.

Q3f. Premature Ejaculation

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 3, Q3a)

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed:
  1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a bothersome fall in latency, often to **≤3 minutes** (acquired).
  2. Inability to delay ejaculation on all or nearly all penetrations.
  3. Negative personal consequences: distress, frustration, avoidance.
- **Types:** lifelong, acquired, variable and subjective.

Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE (premature ejaculation):** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.
- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials (population median 5.4 minutes).
- **PEDT (Premature Ejaculation Diagnostic Tool)**; IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor])	30–60 mg, 1–3 h before
Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine spray	At least 5 min before; condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI

- SSRIs raise synaptic serotonin and delay ejaculation. Avoid in bipolar disorder.
- **Also know:** tramadol (addiction risk); high dropout from drug therapy.

Diagram to draw: office algorithm: confirm PE → secondary to ED? → treat cause; else classify lifelong or acquired.

Key points to remember:

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Rapid Revision Sheet – Practice Set 11 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe pzthophysiology of urosepsis and its management. (LAQ)

- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Stent = nephrostomy for drainage; delay definitive stone surgery.

Q2. Describe anatomy of pelvic floor in females and its importance in surgeries described for SUI (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey I suspension, II attachment (ATFP), III fusion (urethral hypermobility).
- Integral theory → mid-urethral sling; hammock → backing, not repositioning.
- Burch → Cooper's ligament; TOT → obturator foramen; PVS → bladder neck.

Q3a. Retrocaval ureter (SAQ)

- Venous anomaly (persistent subcardinal vein); right side; 1 in 1500; male 3:1.
- Type I fishhook obstructs; type II does not.
- CT urography; anterior transposition and ureteroureterostomy.

Q3b. Glandular hypospadias (SAQ)

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

Q3c. Treatment of submeatal stricture urethra in males (SAQ)

- RUG first; adult meatal stenosis is usually the tip of a fossa stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Q3d. TOT in SUI (SAQ)

- Delorme 2001; anchors in obturator muscles.
- TOMUS: equivalent to retropubic (77.7% vs 80.8%).
- Fewer bladder injuries, more groin pain; weaker in ISD.

Q3e. Testicular biopsy (SAQ)

- Indication: azoospermia, normal testes, vasa and FSH.
- Bouin's, never formalin; no forceps; touch prep.
- Haematoma commonest; epididymal biopsy most serious.

Q3f. Premature Ejaculation (SAQ)

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.